

CS214 – Computer Organization

Mid-Semester Examination Syllabus

Syllabus

1. Digital Logic Fundamentals

- Boolean algebra and Boolean functions
- Combinational logic circuits
- Adders and arithmetic circuits
- Multiplexers and decoders
- Propagation delay and hazards

2. Sequential Logic

- Latches and flip-flops
- Registers and shift registers
- Counters
- Timing constraints
- Synchronous sequential circuits
- Finite State Machines (FSMs)
- Moore and Mealy machines
- State diagrams, state tables, state assignment, and implementation

3. Basic Computer Organization

- Basic elements of a computer
- von Neumann model
- Stored-program concept
- Programmer-visible (architectural) state
- Instruction Set Architecture (ISA)
- ISA versus microarchitecture

4. LC-3 Instruction Set Architecture

- LC-3 architecture and programmer-visible state
- Instruction formats and opcodes
- Operate, data movement, and control instructions
- Addressing modes
- Condition codes
- LC-3 assembly language
- Instruction processing cycle: Fetch, Decode, Evaluate Address, Fetch Operands, Execute, and Store Result

5. Microarchitecture

- Microarchitecture fundamentals

- Datapath and control
- Register file, ALU, PC, IR, MAR, and MDR
- Control signals and finite-state control
- Single-cycle microarchitecture
- Multi-cycle microarchitecture
- Comparison of single-cycle and multi-cycle implementations

Recommended References

1. M. Morris Mano and Michael D. Ciletti, *Digital Design*.
2. Yale N. Patt and Sanjay J. Patel, *Introduction to Computing Systems: From Bits & Gates to C/C++ & Beyond*.
3. David Money Harris and Sarah L. Harris, *Digital Design and Computer Architecture*.